

Madhan Senthilkumar

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PROFESSIONAL SUMMARY

AI/ML enthusiast with 2 internships and hands-on experience across classification, regression, clustering, deep learning, computer vision, and LLM-based systems. Deployed projects including FitCheck, a dual-mode Explainable RAG platform for resume-JD matching; a Random Forest-based customer churn prediction platform (81.5% accuracy); and a real-time YOLOv8 traffic analytics system. Proficient in Python, Scikit-learn, TensorFlow, LangChain, OpenCV, Pandas, and SQL, with practical experience across the ML lifecycle – EDA, feature engineering, model tuning, and evaluation. IBM-certified in Python for Data Science.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning – Parul University, Vadodara

CGPA: 8.30 | Aug 2023 – 2027

EXPERIENCE

Data Science Intern

Future Interns | Dec 2025 – Jan 2026

- Designed a weighted scoring model across 5-point Likert scales to quantify student satisfaction, identifying a top-performing metric at 4.21/5 and flagging 3 critical areas below 3.6/5 for institutional review.
- Delivered an interactive visual report using Pandas and Seaborn translating raw survey responses into actionable recommendations for 2 key improvement areas.

Machine Learning Intern

Cognifyz Technologies (Remote) | Nov 2025 – Dec 2025

- Designed and executed end-to-end EDA and preprocessing pipelines using Python and Pandas on large-scale datasets, improving data reliability for downstream modeling.
- Built and evaluated multiple classification and regression models using Scikit-learn, selecting the best-performing model based on accuracy, precision, and F1-score metrics.

PROJECTS

FitCheck – Dual-Mode Explainable RAG Platform [\[GitHub\]](#) [\[Live Demo\]](#) *React, FastAPI, LangChain, ChromaDB, Groq (Llama 3.1)* | 2026

- Built a dual-mode RAG platform enabling candidates to detect cross-JD requirement conflicts and recruiters to auto-rank up to 10 resumes via a source-cited, confidence-scored matching engine.
- Engineered structured LLM-based candidate scoring (match %, why-select/why-not-select rationale) with a fairness-audit layer disclosing sensitive-signal presence without influencing rankings.

Customer Churn Prediction & Retention Platform [\[Live Demo\]](#)

Python, Scikit-learn, SHAP, Streamlit | 2026

- Developed a hyperparameter-tuned Random Forest Classifier (81.5% accuracy, 0.854 ROC-AUC) predicting customer churn risk on 7,000+ Telco records.
- Integrated SHAP TreeExplainer for local feature attribution and built a vectorized batch-prediction engine for bulk CSV risk analysis.

AI-Powered Traffic Analytics System [\[GitHub\]](#)

Python, YOLOv8, ByteTrack, OpenCV, Streamlit, SQLite | 2026

- Built an end-to-end CV pipeline with YOLOv8 detection and ByteTrack tracking to persistently identify vehicles/pedestrians across live video frames.
- Engineered speed estimation and traffic density classification (Low/Medium/Heavy), with rule-based violation detection logged to SQLite and visualized on a live Streamlit dashboard.

SafeGo – Inclusive Ride Assistance & Safety Platform [\[GitHub\]](#)

MERN, Firebase, ML, OSRM | 2025

- Embedded a Random Forest Classifier and Regressor for geospatial safety scoring, integrating ML predictions into a production backend via a Python inference layer.
- Designed real-time route navigation, distance, and ETA prediction pipelines using OpenStreetMap and OSRM data.

TECHNICAL SKILLS

Generative AI & LLM: LangChain, RAG, Vector Databases (ChromaDB), Groq API, Prompt Engineering, Sentence-Transformers

Machine Learning: Classification, Regression, Clustering, Random Forest, Model Evaluation, Hyperparameter Tuning, Feature Engineering

Deep Learning & Computer Vision: TensorFlow, YOLOv8, OpenCV, Object Detection, Multi-Object Tracking (ByteTrack)

Data & Analysis: Pandas, NumPy, EDA, Scikit-learn, SQL, MongoDB, MySQL

Programming: Python, Java, JavaScript, TypeScript, SQL

Tools & Platforms: FastAPI, React, Git, GitHub, Docker, Jupyter Notebook, Vercel, Streamlit

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Developer – Associate (Infosys Springboard)
- Oracle – Agentic AI Certified Foundations Associate
- Anthropic – Claude Code 101
- HackerRank – SQL (Advanced) Certificate
- IBM – Python for Data Science (2025)